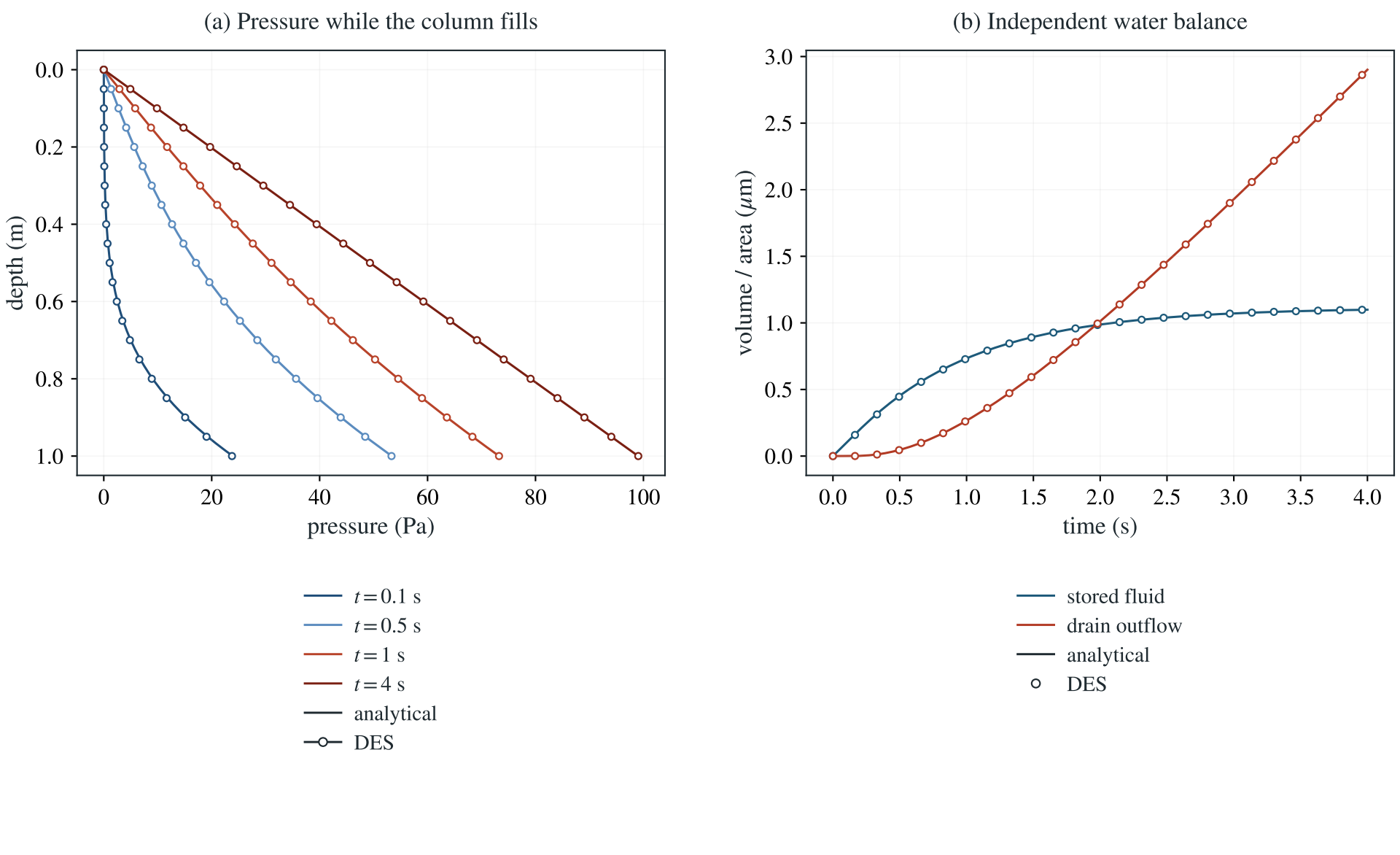

(a) Pressure while the column fills
depth (m)
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100
pressure (Pa)
t = 0.1 s
t = 0.5 s
t = 1 s
t = 4 s
analytical
DES
(b) Independent water balance
volume / area (μm)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
time (s)
stored fluid
drain outflow
analytical
DES